

SUCCESS SECRETS WITH ALEX

Mini-Project: API fetch → save to file

SHERIN MARY P.J

GOAL

- Use **Vibe Coding**
- Talk to a new PUBLIC API that you've not used before.
- Write JS code to get data from the API.
- Save that to a file in replit → .csv file.
- The File Should have Rows and Columns.
- Create a new .csv file and store data
- Add data to an existing File

Prompt to Generate the Project

Purpose

Act as an expert JavaScript and Node.js developer. I want to build a beginner-friendly project that fetches JSON data from a **new Public API**, converts the JSON data into **CSV format**, and saves it as a **.csv** file.

Requirements

Please write a complete JavaScript program that satisfies all of the following conditions:

1. Use a **new Public API** that returns JSON data (do not use any API that has been used before).
2. Use the **fetch()** function with **async** and **await** to retrieve the data.
3. Convert the JSON response into CSV format.
4. Create a new CSV file if it does not already exist.
5. The CSV file should contain properly formatted **rows and columns** with a header row.
6. If the CSV file already exists, append the new data to the existing file instead of overwriting it.
7. Use Node.js built-in **File System (fs)** module for file operations.
8. Include proper error handling using **try...catch**.
9. Print a success message in the console after the CSV file has been created or updated.

Input

- Public API URL
- JSON data returned by the API

Expected Output

- A CSV file (for example, **products.csv**)
- The file should contain rows and columns with meaningful headers.
- On the first execution, the program should create the CSV file and store the data.
- On subsequent executions, the program should append new data to the existing CSV file without deleting the previous data.
- Display a success message such as: **"Data successfully saved to products.csv"** after execution.

Please provide clean, beginner-friendly JavaScript code that can be run directly in Visual Studio Code using Node.js.

CODE

```
const fs = require("fs");

async function fetchProducts() {

  try {

    // Fetch data from the Public API

    const response = await fetch("https://dummyjson.com/products");

    // Convert response to JSON

    const data = await response.json();

    // Get the products array

    const products = data.products;

    // Check if products.csv already exists

    const fileExists = fs.existsSync("products.csv");

    // Create the file with a header if it doesn't exist

    if (!fileExists) {

      fs.writeFileSync(

        "products.csv",

        "ID,Title,Price,Category\n"

      );

    }

  }

}
```

```
// Convert JSON to CSV rows

let csv = "";

products.forEach((product) => {

    csv +=

`${product.id},${product.title},${product.price},${product.category}"\n
`;

});

// Append the new data to the existing CSV file

fs.appendFileSync("products.csv", csv);

console.log("✅ Data successfully saved to products.csv");

} catch (error) {

    console.error("❌ Error:", error.message);

}

}

// Call the function

fetchProducts();
```

OUTPUT

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell + - [ ] [ ] ... | [ ] [ ]
PS C:\Users\acer\Downloads\AI SKILLS\JSON TO CSV> node index.js
CSV file created successfully!
```

	A	B	C	D	E
1	ID	Title	Price	Category	
2		1 Essence M	9.99	beauty	
3		2 Eyeshadow	19.99	beauty	
4		3 Powder Ca	14.99	beauty	
5		4 Red Lipsti	12.99	beauty	
6		5 Red Nail P	8.99	beauty	
7		6 Calvin Kle	49.99	fragrances	
8		7 Chanel Co	129.99	fragrances	
9		8 Dior J'ado	89.99	fragrances	
10		9 Dolce Shir	69.99	fragrances	
11		10 Gucci Blo	79.99	fragrances	
12		11 Annibale C	1899.99	furniture	
13		12 Annibale C	2499.99	furniture	
14		13 Bedside T	299.99	furniture	
15		14 Knoll Saar	499.99	furniture	
16		15 Wooden B	799.99	furniture	
17		16 Apple	1.99	groceries	
18		17 Beef Steak	12.99	groceries	

	A	B	C	D	
18		17 Beef Steak	12.99	groceries	
19		18 Cat Food	8.99	groceries	
20		19 Chicken M	9.99	groceries	
21		20 Cooking C	4.99	groceries	
22		21 Cucumber	1.49	groceries	
23		22 Dog Food	10.99	groceries	
24		23 Eggs	2.99	groceries	
25		24 Fish Steak	14.99	groceries	
26		25 Green Bell	1.29	groceries	
27		26 Green Chil	0.99	groceries	
28		27 Honey Jar	6.99	groceries	
29		28 Ice Cream	5.49	groceries	
30		29 Juice	3.99	groceries	
31		30 Kiwi	2.49	groceries	
32					

Add data to an existing

	A	B	C	D	E
32		1 Essence M	9.99	beauty	
33		2 Eyeshadow	19.99	beauty	
34		3 Powder Ca	14.99	beauty	
35		4 Red Lipstic	12.99	beauty	
36		5 Red Nail P	8.99	beauty	
37		6 Calvin Kle	49.99	fr fragrances	
38		7 Chanel Co	129.99	fr fragrances	
39		8 Dior J'ado	89.99	fr fragrances	
40		9 Dolce Shir	69.99	fr fragrances	
41		10 Gucci Blo	79.99	fr fragrances	
42		11 Annibale C	1899.99	furniture	
43		12 Annibale C	2499.99	furniture	
44		13 Bedside T	299.99	furniture	
45		14 Knoll Saar	499.99	furniture	
46		15 Wooden B	799.99	furniture	
47		16 Apple	1.99	groceries	
48		17 Beef Steak	12.99	groceries	
49		18 Cat Food	8.99	groceries	

[FINAL STEP] Instructions for Sharing Your "Mini-Project: API fetch → save to file" Google Doc

Congratulations on completing your "Mini-Project: API fetch → save to file"! Sharing your work with our community is a powerful step toward turning them into reality. Here's how you can share your document with view-only access:

Add your Prompt, Code, Output and Screenshots to the Google Doc.

Steps to Share Your Google Doc:

- 1. Open Your Document:**
 - Navigate to your Google Drive and open the document you've created.
- 2. Access Sharing Settings:**
 - Click on the blue "Share" button located in the top-right corner of the document.
- 3. Get Shareable Link:**
 - In the sharing window, click on "Get shareable link" in the top-right corner.
- 4. Set Link Permissions:**
 - A link will be generated, and you'll see a dropdown menu.
 - Select "Anyone with the link can view" to ensure others can see your document but not edit it.
- 5. Copy the Link:**
 - Click "Copy link" to copy the URL to your clipboard.
- 6. Share with the Community:**
 - Paste the link into our Skool Community to share your document with fellow members.
- 7. Add your Name on Top:**
 - Make sure to mention your name on top so that we know who submitted the assignment.
- 8. Nothing Else:**

- Don't share PDFs, MS Office docs, etc.